Assignment 5

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Please note that

- **Released date: 7th Jul, Fri.**
- **Due date: 14th Jul, Fri, by 11:59pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "122010XXX ASS5.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Problem 1

Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear operator. If $L\left((1,2)^T\right) = (-2,3)^T$ and $L\left((1,-1)^T\right) = (5,2)^T$ find the value of $L\left((2,13)^T\right)$.

Solution

Let

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 2 \\ 13 \end{pmatrix}$$

To determine $L(\mathbf{x})$ we must first express $\mathbf{x}$ as a linear combination

$$\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$$

To do this we must solve the system $U\mathbf{c} = \mathbf{x}$ for $\mathbf{c}$. The solution is $\mathbf{c} = (5,-3)^T$ and it follows that

$$L(\mathbf{x}) = L\left(5\mathbf{u}_1 - 3\mathbf{u}_2\right) = 5L\left(\mathbf{u}_1\right) - 3L\left(\mathbf{u}_2\right) = 5\begin{pmatrix} -2 \\ 3 \end{pmatrix} - 3\begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} -25 \\ 9 \end{pmatrix}$$

Problem 2

Determine whether the following are linear transformations from $\mathbb{R}^2$ into $\mathbb{R}^3$.

(a) $L(\mathbf{x}) = (x_1, x_2, 2)^T$

(b) $L(\mathbf{x}) = (x_1, x_2 - x_1, x_1 + 2x_2)^T$

Solution

(a) $L(\mathrm{x})$ is not a linear transformation, since

$$L(\alpha\mathrm{x}) = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ 2 \end{pmatrix}, \alpha L(\mathrm{x}) = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ 2\alpha \end{pmatrix}$$

it follows that $L(\alpha\mathrm{x}) \neq \alpha L(\mathrm{x})$

(b) $L(\mathrm{x})$ is a linear transformation, since

$$L(\alpha\mathbf{x}+\beta\mathbf{y}) = \begin{pmatrix} \alpha x_1+\beta y_1 \\ \alpha x_2+\beta y_2-(\alpha x_1+\beta y_1) \\ \alpha x_1+\beta y_1+2(\alpha x_2+\beta y_2) \end{pmatrix}$$
$$= \alpha \begin{pmatrix} x_1 \\ x_2-x_1 \\ x_1+2x_2 \end{pmatrix} + \beta \begin{pmatrix} y_1 \\ y_2-y_1 \\ y_1+2y_2 \end{pmatrix} = \alpha L(\mathbf{x})+\beta L(\mathbf{y})$$

Problem 3

Determine whether the following are linear transformations from P_2 to P_3.

(a) $L(p(x)) = xp(x)$

(b) $L(p(x)) = x^2 + p(x)$

Solution

Let $p_1(x) = ax+b$ is a polynomial in P_2, and $p_2(x) = cx+d$ is also a polynomial in P_2, we assume α and β are scalars.

(a)

$$\begin{aligned} L(\alpha p_1(x)+\beta p_2(x)) &= x(\alpha p_1(x)+\beta p_2(x)) \\ &= x\alpha p_1(x)+x\beta p_2(x) \\ &= \alpha L(p_1(x))+\beta L(p_2(x)) \\ &= \alpha a x^2+\alpha b x+\beta c x^2+\beta d x \\ &= (\alpha a+\beta c)x^2+(\alpha c+\beta d)x \end{aligned}$$

Thus, L is a linear transformation from P_2 to P_3.

(b)

$$\begin{aligned} L(\alpha p_1(x)+\beta p_2(x)) &= x^2+\alpha p_1(x)+\beta p_2(x) \\ \alpha L(p_1(x))+\beta L(p_2(x)) &= \alpha x^2+\alpha p_1(x)+\beta x^2+\beta p_2(x) \\ &= (\alpha+\beta)x^2+\alpha p_1(x)+\beta p_2(x) \\ L(\alpha p_1(x)+\beta p_2(x)) &\neq \alpha L(p_1(x))+\beta L(p_2(x)) \end{aligned}$$

Thus, L is not a linear transformation from P_2 to P_3.

Problem 4

For each $f \in C[0,1]$, define $L(f) = F$, where $F(x) = \int_0^x f(t)dt,\ 0 \le x \le 1$. Show that L is a linear operator on $C[0,1]$ and then find $L(e^x)$ and $L(x^2)$.

Solution

If $f, g \in C[0,1]$ then

$$\begin{aligned} L(\alpha f+\beta g) &= \int_0^x (\alpha f(t)+\beta g(t))dt \\ &= \alpha\int_0^x f(t)dt+\beta\int_0^x g(t)dt \\ &= \alpha L(f)+\beta L(g) \end{aligned}$$

Thus L is a linear transformation from $C[0,1]$ to $C[0,1]$.

$$L\left(e^x\right)=\int_0^x \left(e^t\right)dt = e^x-1$$
$$L\left(x^2\right)=\int_0^x \left(t^2\right)dt = \frac{1}{3}x^3$$

Problem 5

Let

$$\mathbf{y}_1=\begin{pmatrix}1\\1\\1\end{pmatrix},\quad \mathbf{y}_2=\begin{pmatrix}1\\1\\0\end{pmatrix},\quad \mathbf{y}_3=\begin{pmatrix}1\\0\\0\end{pmatrix}$$

and let $\mathcal{I}$ be the identity operator on $\mathbb{R}^3$.

(a) Find the coordinates of $\mathcal{I}(\mathbf{e}_1), \mathcal{I}(\mathbf{e}_2)$, and $\mathcal{I}(\mathbf{e}_3)$ with respect to $\{\mathbf{y}_1,\mathbf{y}_2,\mathbf{y}_3\}$.

(b) Find a matrix A such that $A\mathbf{x}$ is the coordinate vector of $\mathbf{x}$ with respect to $\{\mathbf{y}_1,\mathbf{y}_2,\mathbf{y}_3\}$.

Solution

(a)
$$\begin{pmatrix}1\\0\\0\end{pmatrix}=\mathcal{I}(\mathbf{e}_1)=0\begin{pmatrix}1\\1\\1\end{pmatrix}+0\begin{pmatrix}1\\1\\0\end{pmatrix}+1\begin{pmatrix}1\\0\\0\end{pmatrix}=0\mathbf{y}_1+0\mathbf{y}_2+1\mathbf{y}_3$$
$$\begin{pmatrix}0\\1\\0\end{pmatrix}=\mathcal{I}(\mathbf{e}_2)=0\begin{pmatrix}1\\1\\1\end{pmatrix}+1\begin{pmatrix}1\\1\\0\end{pmatrix}+(-1)\begin{pmatrix}1\\0\\0\end{pmatrix}=0\mathbf{y}_1+1\mathbf{y}_2-1\mathbf{y}_3$$
$$\begin{pmatrix}0\\0\\1\end{pmatrix}=\mathcal{I}(\mathbf{e}_1)=1\begin{pmatrix}1\\1\\1\end{pmatrix}+(-1)\begin{pmatrix}1\\1\\0\end{pmatrix}+0\begin{pmatrix}1\\0\\0\end{pmatrix}=1\mathbf{y}_1-1\mathbf{y}_2+0\mathbf{y}_3$$

(b)
$$A=\begin{pmatrix}0&0&1\\0&1&-1\\1&-1&0\end{pmatrix}$$

Problem 6

Determine the kernel of each of the following linear operators on $\mathbb{R}^3$:

(a)
$$L(\boldsymbol{x})=\begin{bmatrix}x_3 & x_2 & x_1\end{bmatrix}^T$$

(b)
$$L(\boldsymbol{x})=\begin{bmatrix}x_1 & x_2 & 0\end{bmatrix}^T$$

(c)

$$L(\boldsymbol{x}) = \begin{bmatrix} x_1 & x_1 & x_1 \end{bmatrix}^T$$

Solution

(a)

$$L(\boldsymbol{x}) = \begin{bmatrix} x_3 & x_2 & x_1 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

$$\ker(L) = \{\mathbf{0}\}.$$

(b)

$$L(\boldsymbol{x}) = \begin{bmatrix} x_1 & x_2 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

$$\ker(L) = \text{Span}\{\boldsymbol{e}_3\}.$$

(c)

$$L(\boldsymbol{x}) = \begin{bmatrix} x_1 & x_1 & x_1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{0}.$$

$$\ker(L) = \text{Span}\{\boldsymbol{e}_2, \boldsymbol{e}_3\}.$$

Problem 7

Let $\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ and $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ be ordered bases for $\mathbb{R}_2$, where

$$\boldsymbol{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{u}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Let L be the linear transformation defined by $L(\boldsymbol{x}) = \begin{bmatrix} -x_1 & x_2 \end{bmatrix}^T$, and let $\boldsymbol{B}$ be the matrix representing L with respect to $\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$.
(a) Find the transition matrix $\boldsymbol{S}$ corresponding to the change of basis from $\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ to $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.
(b) Find the matrix $\boldsymbol{A}$ representing L with respect to $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ by computing $\boldsymbol{SBS}^{-1}$.
(c) Verify that

$$L(\boldsymbol{v}_1) = a_{11}\boldsymbol{v}_1 + a_{21}\boldsymbol{v}_2, \qquad L(\boldsymbol{v}_2) = a_{12}\boldsymbol{v}_1 + a_{22}\boldsymbol{v}_2.$$

Solution

(a)

$$\boldsymbol{U} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{V} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

$$\boldsymbol{V}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}, \quad \boldsymbol{S} = \boldsymbol{V}^{-1}\boldsymbol{U} = \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}.$$

(b)

$$L(\boldsymbol{u}_1) = (-1, 1)^T = 0 + \boldsymbol{u}_2, \quad L(\boldsymbol{u}_2) = (1, 1)^T = \boldsymbol{u}_1 + 0.$$

$$\boldsymbol{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\boldsymbol{A} = \boldsymbol{SBS}^{-1} = \frac{-1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -4 & -1 \end{bmatrix}.$$

(c)

$$L(\boldsymbol{v}_1) = (-2,1)^T = 1\cdot(2,1)^T + (-4)\cdot(1,0)^T = 1\cdot\boldsymbol{v}_1 + (-4)\cdot\boldsymbol{v}_2,$$
$$L(\boldsymbol{v}_2) = (-1,0)^T = 0\cdot(2,1)^T + (-1)\cdot(1,0)^T = 0\cdot\boldsymbol{v}_1 + (-1)\cdot\boldsymbol{v}_2.$$

Hence

$$L(\boldsymbol{v}_1) = a_{11}\boldsymbol{v}_1 + a_{21}\boldsymbol{v}_2, \qquad L(\boldsymbol{v}_2) = a_{12}\boldsymbol{v}_1 + a_{22}\boldsymbol{v}_2.$$

Problem 8

Let

$$b_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad b_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad b_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

and let L be the linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ defined by

$$L(x) = x_1 b_1 + x_2 b_2 + (x_1 + x_2)\, b_3.$$

Find the matrix A representing L with respect to the standard basis $\{\mathrm{e1}, \mathrm{e2}\}$ of $\mathbb{R}^2$ and the basis $\{b1, b2, b3\}$ of $\mathbb{R}^3$.

Solution

$$[L(e1)]_{\mathcal{B}} = b_1 + b_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

$$[L(e2)]_{\mathcal{B}} = b_2 + b_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Problem 9

Let L be the linear transformation mapping $\mathbb{R}^3$ into $\mathbb{R}^2$ defined by

$$L(\boldsymbol{x}) = x_1 b_1 + (x_2 - x_3)\, b_2$$

for each $x \in \mathbb{R}^3$, where

$$b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } b_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Find the matrix A representing L with respect to the ordered bases $\{e_1, e_2, e_3\}$ and $\{b_1, b_2\}$

Solution

$$L(e_1) = 1b_1 + 0b_2$$
$$L(e_2) = 0b_1 + 1b_2$$
$$L(e_3) = 0b_1 - 1b_2$$

The ith column of A is determined by the coordinates of $L(e_i)$ with respect to $\{b_1, b_2\}$ for $i = 1, 2, 3$. Thus $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$.

Problem 10

Find the matrix representation of the linear transformation $T(f(t)) = f(3t-2)$ from $\mathbb{P}_2$ to $\mathbb{P}_2$, where $\mathbb{P}_2 = \left\{a_0 + a_1 x + a_2 x^2 \mid a_0, a_1, a_2 \in \mathbb{R}\right\}$.

Solution

A basis for $\mathbb{P}_2$ is : $\left[1, t, t^2\right]$.

$$T(1) = 1 = 1 \cdot 1 + 0 \cdot t + 0 \cdot t^2$$
$$T(t) = 3t - 2 = -2 \cdot 1 + 3 \cdot t + 0 \cdot t^2$$
$$T\left(t^2\right) = (3t-2)^2 = 4 \cdot 0 - 12 \cdot t + 9t^2$$

the matrix of T w.r.t to $\mathbb{P}_2$ is

$$B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & -12 \\ 0 & 0 & 9 \end{bmatrix}$$

Problem 11

Let L be the linear tranformation mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ defined by $L(x) = Ax$, where

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

and let

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Find the transition matrix V corresponding to a change of basis from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, and use it to determine the matrix B representing L with respect to $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$

Solution

$$V=\begin{bmatrix}1&1&0\\1&2&-2\\1&0&1\end{bmatrix}.V^{-1}=\begin{bmatrix}-2&1&2\\3&-1&-2\\2&-1&-1\end{bmatrix}$$

$$B=V^{-1}AV=\begin{bmatrix}-2&1&2\\3&-1&-2\\2&-1&-1\end{bmatrix}\begin{bmatrix}3&-1&-2\\2&0&-2\\2&-1&-1\end{bmatrix}\begin{bmatrix}1&1&0\\1&2&-2\\1&0&1\end{bmatrix}=\begin{bmatrix}0&0&0\\0&1&0\\0&0&1\end{bmatrix}.$$

Problem 12

Let L be the linear operator on $\mathbb{R}^3$ defined by

$$L(\mathbf{x})=\begin{pmatrix}2x_1-x_2-x_3\\2x_2-x_1-x_3\\2x_3-x_1-x_2\end{pmatrix}$$

Determine the standard matrix representation A of L, and use A to find $L(\mathbf{x})$ for each of the following vectors $\mathbf{x}$: (a) $\mathbf{x}=(1,1,1)^T$ (b) $\mathbf{x}=(2,1,1)^T$ (c) $\mathbf{x}=(-5,3,2)^T$

Solution

$$L(\vec{x})=A\vec{x},\quad A=\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}$$

(a) $L(\vec{x})=A\vec{x}=\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$

(b) $L(\vec{x})=A\vec{x}=\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}\begin{pmatrix}2\\1\\1\end{pmatrix}=\begin{pmatrix}2\\-1\\-1\end{pmatrix}$

(C) $L(\vec{x})=A\vec{x}=\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix}\begin{pmatrix}-5\\3\\2\end{pmatrix}=\begin{pmatrix}-15\\9\\6\end{pmatrix}$

Problem 13

Show that the transformation T defined by

$$T\left(x_1,x_2\right)=\left(x_1-4\left|x_2\right|,x_1-4x_2\right)$$

is not linear.

Solution

We have

$$T(0,1)+T(0,-1)=(-4,-4)+(-4,4)=(-8,0)$$

but

$$T((0,1)+(0,-1)) = T(0,0) = (0,0)$$

These are not equal, so T does not preserve vector addition. Therefore it is not a linear transformation.